CARTER GRIMMEISEN

SOFTWARE ENGINEER

CONTACT

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EDUCATION

MS COMPUTER SCIENCE
Expected 2027 - 4.0 GPA
University of Alabama in Huntsville

BS COMPUTER SCIENCE
May 2022 - 4.0 GPA
University of Alabama in Huntsville

SKILLS

LANGUAGES
Python, TypeScript, Ada, Rust, C++, Bash, Java, C#, Kotlin, C, HTML/CSS/JS

FRAMEWORKS AND TOOLS
React Native, Flutter, Vue.js, SvelteKit, Node.js, PyTorch, PostgreSQL, SQLite/DuckDB, Docker, CI/CD, Git, Express.js, Flask, Plotly Dash, MPI, React, Svelte, Electron, GitLab, GitHub, Docker Compose, MongoDB, Conda, Cython

AI-ASSISTED DEVELOPMENT
Claude Code, GitHub Copilot, Cursor, agent skills and workflows, self-hosted open models

CERTIFICATIONS

Linux+
LPIC-1 Linux Administrator
Security+
Network+

Active DoD Secret clearance

EXPERIENCE

PRINCIPAL SOFTWARE ENGINEER — MAY 2026 - PRESENT

TECHNICAL CONSULTING SOLUTIONS
Subcontractor to Northrop Grumman
+ Develop software for a large, fielded Ada codebase as part of a 10-12 person team
+ Built a web-based log analysis tool that processes run logs as they arrive on the log server, giving developers and testers visibility into system behavior and exposing a class of slowdowns behind timeouts and failures
+ Own vulnerability scanning and metrics reporting for the codebase

LEAD SOFTWARE ENGINEER — JUNE 2022 - MAY 2026

CFD RESEARCH CORP.

LEADERSHIP
+ Led a 3-person team within an 8-10 person engineering group; conducted performance reviews, mentored engineers, interviewed candidates, and matched staff to projects
+ Served as technical lead on greenfield SBIR projects ($800K-$6M) for scientific customers, owning requirements, planning, delivery, and customer reviews across mobile, web, ML, and DevOps work
+ Authored the technical approach (architecture, design, implementation plan) for ~6 proposals and presented it in customer Q&A; 3 won outright and 1 advanced as a joint effort, and the resulting expertise supported a separate ~$10M award
+ Managed customer scope and expectations by negotiating phased delivery, proposing alternatives when approaches fell short, and communicating revised timelines when schedules slipped

DELIVERY
+ Reduced ML training time from ~8 hours to under 30 minutes (20x throughput) by moving a 200K-sample dataset into SQLite/DuckDB and implementing a custom PyTorch dataloader
+ Designed and built a web-based simulation platform for ~12 scientists, replacing manual script-and-CSV workflows with run configuration, job monitoring, and results visualization (Postgres job queue, custom scheduler, role-based access control); automated CI/CD so scientists could release code and configuration updates independently
+ Led development of an offline-first React Native mobile app with cloud sync and 2D/3D visualization (VTK.js) for blast overpressure exposure, used by soldiers and trainers across dozens of devices
+ Led a cross-functional team of scientists and developers to parallelize, package, and document a space-weather simulation toolkit for HPC deployment
+ Developed the real-time monitoring and configuration web UI (SvelteKit) for a spin-out company's software-defined firewall product
+ Built a Python application packager and custom Python distribution builder to streamline licensing, compilation, and deployment of internal applications and libraries (Python, Docker)
+ Delivered 3D visualization and web scraping projects for scientist teams
+ Developed agentic coding workflows (reusable skills, coding standards, setup documentation) and deployed self-hosted open models for cost-sensitive and air-gapped environments

SOFTWARE ENGINEER, INTERN — MAY 2020 - MAY 2022

NORTHROP GRUMMAN
+ Automated an ~18-hour biweekly vulnerability-scan and metrics reporting process, eliminating recurring engineering effort
+ Developed an Ada mocking framework to enable unit testing of a legacy codebase on a 7-8 person DevOps team
+ Maintained build and CI/CD pipelines for the team

SOFTWARE DEVELOPER — MARCH 2017 - MAY 2020

AVILUTION, LLC.

+ Owned architecture, deployment, and support as one of two engineers at an early-stage avionics startup
+ Built a drag-and-drop display designer (Electron, Vue.js, TypeScript) with dozens of configurable widgets and aircraft configuration management, a key product selling point
+ Implemented ECDSA signing for flight-critical configurations and contributed to core avionics software in C
+ Developed low-level hardware drivers and a custom embedded Yocto build for an avionics-as-software platform

OPEN SOURCE

ZOD-PRISMA — https://www.npmjs.com/package/zod-prisma

MADE WITH TYPESCRIPT, PRISMA, ZOD

Open-source library that generates Zod validation schemas from Prisma data models. **900+ GitHub stars**, **2.3M lifetime downloads**.